

## PADTHAWAY SOUTH, Australia

WMO: 958230

Lat: 36.6539S

Lon: 140.5211E

Elev: 37

StdP: 100.88

Time Zone: 9.50 (AUA)

Period: 03-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 0.5        | 1.8 | -2.2                          | 3.2 | 19.6 | -0.6 | 3.6 | 12.9 | 10.9                  | 11.9 | 10.1 | 12.9 | 0.9                   | 350  | 0.527 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 15.9                   | 37.0            | 18.8 | 34.4 | 18.2 | 32.1 | 17.6 | 20.8                | 29.8 | 19.8 | 29.2 | 18.9 | 28.4 | 5.0                  | 340  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 18.5                            | 13.4 | 22.4 | 17.1 | 12.2 | 21.8 | 15.9 | 11.3 | 20.8 | 60.2          | 29.5 | 56.7 | 29.1 | 53.7 | 28.4 | 24.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 9.4           | 8.4  | 7.6 | -1.7                       | 41.8 | 1.1                | 1.7 | -2.5                                               | 43.0 | -3.1       | 44.0 | -3.7       | 44.9 | -4.5       | 46.1 |    |    |
| (5)               |      |     | -1.9                       | 22.5 | 1.1                | 0.8 | -2.7                                               | 23.2 | -3.4       | 23.6 | -4.0       | 24.1 | -4.8       | 24.7 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 14.7  | 20.4  | 20.1  | 18.0  | 14.9  | 12.2  | 9.9   | 9.7   | 10.3  | 11.9  | 14.0  | 16.5  | 18.5 | (6)  |
| (7)  |                                                                                  | DBStd       | 5.23  | 4.40  | 4.23  | 4.21  | 3.49  | 2.92  | 2.40  | 2.21  | 2.53  | 3.01  | 4.00  | 4.46  | 4.29 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 119   | 0     | 0     | 0     | 1     | 10    | 31    | 34    | 27    | 10    | 4     | 1     | 0    | (8)  |
| (9)  |                                                                                  | HDD18.3     | 1673  | 23    | 26    | 58    | 114   | 192   | 254   | 269   | 250   | 196   | 149   | 91    | 51   | (9)  |
| (10) |                                                                                  | CDD10.0     | 1819  | 322   | 282   | 249   | 149   | 78    | 28    | 24    | 36    | 66    | 127   | 195   | 264  | (10) |
| (11) |                                                                                  | CDD18.3     | 331   | 87    | 74    | 49    | 12    | 2     | 0     | 0     | 0     | 2     | 13    | 35    | 56   | (11) |
| (12) |                                                                                  | CDH23.3     | 4887  | 1326  | 1002  | 681   | 162   | 6     | 0     | 0     | 1     | 19    | 212   | 546   | 933  | (12) |
| (13) | CDH26.7                                                                          | 2355        | 705   | 496   | 310   | 48    | 1     | 0     | 0     | 0     | 1     | 77    | 246   | 472   | (13) |      |
| (14) | Wind                                                                             | WSAvg       | 3.7   | 3.8   | 3.6   | 3.4   | 3.0   | 3.3   | 3.3   | 4.0   | 3.8   | 3.9   | 3.8   | 3.8   | (14) |      |
| (15) | Precipitation                                                                    | PrecAvg     | 496   | 25    | 21    | 20    | 29    | 45    | 65    | 70    | 61    | 57    | 43    | 29    | 31   | (15) |
| (16) |                                                                                  | PrecMax     | 670   | 79    | 56    | 57    | 71    | 110   | 108   | 120   | 121   | 117   | 101   | 68    | 113  | (16) |
| (17) |                                                                                  | PrecMin     | 306   | 2     | 3     | 2     | 2     | 6     | 12    | 29    | 7     | 12    | 8     | 7     | 5    | (17) |
| (18) |                                                                                  | PrecStd     | 90    | 19    | 14    | 13    | 18    | 22    | 26    | 24    | 30    | 27    | 26    | 17    | 28   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 41.4  | 38.0  | 37.0  | 31.6  | 23.7  | 18.5  | 17.6  | 21.8  | 25.8  | 32.8  | 34.8  | 39.0 | (19) |
| (20) |                                                                                  |             | MCWB  | 19.9  | 19.3  | 18.3  | 17.2  | 13.9  | 12.6  | 12.2  | 13.1  | 14.9  | 16.1  | 18.4  | 18.5 | (20) |
| (21) |                                                                                  | 2%          | DB    | 37.4  | 35.1  | 33.3  | 27.4  | 21.0  | 16.4  | 15.7  | 18.8  | 22.5  | 28.6  | 32.7  | 35.6 | (21) |
| (22) |                                                                                  |             | MCWB  | 19.3  | 18.9  | 18.2  | 15.6  | 13.7  | 12.1  | 11.5  | 12.3  | 13.3  | 15.0  | 17.1  | 17.6 | (22) |
| (23) |                                                                                  | 5%          | DB    | 33.8  | 32.7  | 30.3  | 24.8  | 18.9  | 15.2  | 14.5  | 16.3  | 19.9  | 25.2  | 29.7  | 32.0 | (23) |
| (24) |                                                                                  |             | MCWB  | 18.5  | 18.6  | 17.2  | 14.7  | 13.3  | 11.7  | 11.1  | 11.4  | 12.6  | 14.4  | 16.9  | 16.9 | (24) |
| (25) | 10%                                                                              | DB          | 30.4  | 29.6  | 27.0  | 22.1  | 17.2  | 14.2  | 13.6  | 14.7  | 17.7  | 22.4  | 26.2  | 28.5  | (25) |      |
| (26) |                                                                                  | MCWB        | 17.6  | 17.8  | 16.3  | 14.3  | 12.9  | 11.3  | 10.6  | 10.9  | 12.1  | 13.8  | 15.8  | 16.4  | (26) |      |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 22.0  | 22.6  | 20.4  | 18.1  | 16.1  | 14.3  | 13.4  | 14.2  | 15.5  | 18.0  | 20.7  | 21.3 | (27) |
| (28) |                                                                                  |             | MCDWB | 31.3  | 33.4  | 28.3  | 25.3  | 19.1  | 16.4  | 15.4  | 19.2  | 22.5  | 26.4  | 29.3  | 29.4 | (28) |
| (29) |                                                                                  | 2%          | WB    | 20.9  | 20.6  | 19.2  | 17.0  | 15.1  | 13.0  | 12.4  | 13.0  | 14.4  | 16.3  | 19.2  | 19.2 | (29) |
| (30) |                                                                                  |             | MCDWB | 30.3  | 29.9  | 28.4  | 23.0  | 18.3  | 15.2  | 14.5  | 16.9  | 20.0  | 24.4  | 27.8  | 28.8 | (30) |
| (31) |                                                                                  | 5%          | WB    | 19.7  | 19.6  | 18.3  | 16.0  | 14.2  | 12.2  | 11.7  | 12.1  | 13.5  | 15.3  | 17.9  | 18.2 | (31) |
| (32) |                                                                                  |             | MCDWB | 30.6  | 28.4  | 27.1  | 21.8  | 17.3  | 14.4  | 13.8  | 15.5  | 18.2  | 23.1  | 26.7  | 28.3 | (32) |
| (33) | 10%                                                                              | WB          | 18.6  | 18.4  | 17.4  | 15.1  | 13.4  | 11.6  | 11.0  | 11.3  | 12.5  | 14.3  | 16.6  | 17.2  | (33) |      |
| (34) |                                                                                  | MCDWB       | 28.9  | 27.0  | 25.7  | 20.4  | 16.5  | 13.7  | 13.0  | 14.2  | 16.8  | 21.0  | 24.2  | 27.0  | (34) |      |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 15.9  | 15.3  | 14.5  | 13.1  | 9.8   | 8.8   | 7.8   | 8.7   | 10.3  | 13.0  | 14.3  | 15.4 | (35) |
| (36) |                                                                                  |             | MCDBR | 21.4  | 19.2  | 18.7  | 16.9  | 11.6  | 9.5   | 8.2   | 10.6  | 13.5  | 18.0  | 19.1  | 21.9 | (36) |
| (37) |                                                                                  |             | MCWBR | 6.4   | 6.2   | 6.2   | 7.1   | 6.3   | 6.2   | 5.4   | 6.0   | 6.3   | 7.5   | 6.8   | 6.8  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 16.6  | 15.4  | 15.4  | 13.3  | 9.3   | 8.2   | 7.0   | 9.4   | 11.5  | 15.1  | 16.2  | 16.4 | (38) |
| (39) |                                                                                  |             | MCWBR | 6.2   | 5.9   | 6.0   | 6.6   | 5.8   | 5.8   | 5.0   | 5.8   | 6.2   | 7.1   | 6.6   | 6.2  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.342 | 0.335 | 0.319 | 0.313 | 0.302 | 0.298 | 0.299 | 0.313 | 0.332 | 0.348 | 0.338 | 0.334 | (40) |      |
| (41) |                                                                                  | taud        | 2.476 | 2.512 | 2.555 | 2.555 | 2.573 | 2.570 | 2.562 | 2.503 | 2.440 | 2.396 | 2.463 | 2.494 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 992   | 976   | 954   | 900   | 852   | 826   | 846   | 885   | 923   | 951   | 990   | 1004  | (42) |      |
| (43) |                                                                                  | Edh at Noon | 116   | 107   | 95    | 83    | 71    | 67    | 71    | 86    | 105   | 119   | 117   | 115   | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 7.63  | 6.70  | 5.28  | 3.72  | 2.50  | 2.05  | 2.20  | 3.08  | 4.24  | 5.62  | 6.70  | 7.48  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.40  | 0.34  | 0.21  | 0.22  | 0.13  | 0.15  | 0.10  | 0.16  | 0.34  | 0.41  | 0.40  | 0.38  | (45) |      |

## Historical Trends

|      |                         | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (18 neighbors) | +0.37   | N/A    | N/A     | N/A   | N/A   | N/A         | -101    | +137    | +41     |

Nomenclature: See separate page